

Week 1 — Introductory Machine Learning Assignment

Student Performance Data Analysis Report
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1. Introduction

This report documents the end-to-end machine learning workflow applied to the UCI Student Performance dataset. The objective was to predict whether a student would **Pass** or **Fail** their final exam (grade $G3 \geq 10$ = Pass) using demographic, social, and academic features. Two supervised classification algorithms — **Logistic Regression** and a **Decision Tree** — were trained and evaluated.

2. Dataset Overview

Attribute	Detail
Source	UCI / Kaggle — student_data.csv
Total Records	395 students
Total Features	33 columns (32 input + 1 target G3)
Feature Types	16 numerical, 17 categorical (object)
Target Variable	pass (1 = $G3 \geq 10$, 0 = $G3 < 10$)
Class Distribution	265 Pass (67.1%) 130 Fail (32.9%)

3. Data Preprocessing

3.1 Missing Value Handling

A check confirmed **zero missing values** across all 33 columns. As a precaution, median imputation logic was applied to numerical columns (`df[col].fillna(median)`) to ensure robustness for future data.

3.2 Target Variable Engineering

A binary target column **pass** was created from the continuous final grade G3: `pass = 1` if $G3 \geq 10$, else `pass = 0`. This converts the regression problem into a binary classification task.

3.3 Categorical Encoding

All 17 object-type columns were encoded using **LabelEncoder**. Encoded features include: school, sex, address, famsize, Pstatus, Mjob, Fjob, reason, guardian, schoolsup, famsup, paid, activities, nursery, higher, internet, romantic.

3.4 Train / Test Split & Normalization

Split	Samples	Proportion
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Training Set	316	80%
Test Set	79	20%

Stratified splitting (stratify=y) was used to preserve the class ratio in both sets. All 32 features were then scaled using **StandardScaler** (zero mean, unit variance) to ensure fair convergence for Logistic Regression.

4. Exploratory Data Analysis (EDA)

- Key visualizations generated during EDA:
- **Pass/Fail Distribution (Count Plot):** 265 students passed (67%) vs 130 failed (33%), showing a moderate class imbalance.
 - **Final Grade Histogram:** G3 grades range 0–20 with a mean of ~10.4. A clear bi-modal shape: one cluster near 0 (zeros) and another around 10–14.
 - **G1 vs G2 Scatter Plot (by Pass/Fail):** Strong positive correlation between first and second period grades. Students who scored above ~9 in both G1 and G2 almost always passed.
 - **Correlation Heatmap:** G1 and G2 are highly correlated with G3 (target). Failures show negative correlation with final grade. Alcohol consumption (Walc/Dalc) shows mild negative correlation with performance.
- Key Insight:** Prior grades (G1, G2) are the strongest predictors of the final grade. Social/behavioral factors (absences, alcohol) have weaker but notable effects.

5. Model Training

Model 1 — Logistic Regression

A linear classifier with a sigmoid activation. Trained on the scaled feature matrix with max_iter=1000 and random_state=42 to ensure reproducibility. Logistic Regression is interpretable and serves as a strong baseline for binary classification.

Model 2 — Decision Tree Classifier

A non-linear tree-based classifier. max_depth=5 was set to prevent overfitting (limiting the depth regularises the model and improves generalisation). random_state=42 ensures reproducibility.

6. Model Evaluation Results

Models were evaluated on the held-out test set (79 samples). Metrics: Accuracy, Precision, Recall, and F1-Score.

6.1 Metrics Summary

Metric	Logistic Regression	Decision Tree
Accuracy	86.08%	88.61%
Precision	95.65%	94.00%

Recall	83.02%	88.68%
F1-Score	88.89%	91.26%

6.2 Per-Class Classification Reports

Logistic Regression

Class	Precision	Recall	F1-Score	Support
Fail	0.73	0.92	0.81	26
Pass	0.96	0.83	0.89	53
Macro Avg	0.84	0.88	0.85	79
Weighted Avg	0.88	0.86	0.86	79

Decision Tree

Class	Precision	Recall	F1-Score	Support
Fail	0.79	0.88	0.84	26
Pass	0.94	0.89	0.91	53
Macro Avg	0.87	0.89	0.87	79
Weighted Avg	0.89	0.89	0.89	79

7. Comparison & Conclusion

Both models achieved strong results on the Student Performance dataset. The **Decision Tree** outperformed Logistic Regression on most metrics, likely because it can capture non-linear interactions between features (e.g., students who failed before AND drink heavily are much more likely to fail again).

- **Decision Tree wins** overall: higher accuracy (88.61% vs 86.08%), better recall for the Fail class (88% vs 92%), and a superior F1-Score (91.26% vs 88.89%).
- **Logistic Regression** has slightly higher precision for Pass (96% vs 94%), making it conservative — it rarely mis-classifies a failing student as passing.
- **Both models handle class imbalance reasonably well** without any oversampling, thanks to stratified splitting and the natural feature strength of G1 and G2.
- **Key predictor:** prior grades (G1, G2) dominate prediction, as seen in the EDA correlation heatmap. Future work could explore feature engineering on these.

8. Future Improvements

- Try **Random Forest** or **Gradient Boosting (XGBoost)** for higher accuracy.
- Apply **SMOTE** or class-weight balancing to further reduce Fail class miss-rate.
- Perform **cross-validation** (k-fold) for more reliable metric estimates.
- Use **feature importance plots** from Decision Tree to identify top predictors.
- Explore **hyperparameter tuning** (GridSearchCV) for both models.

